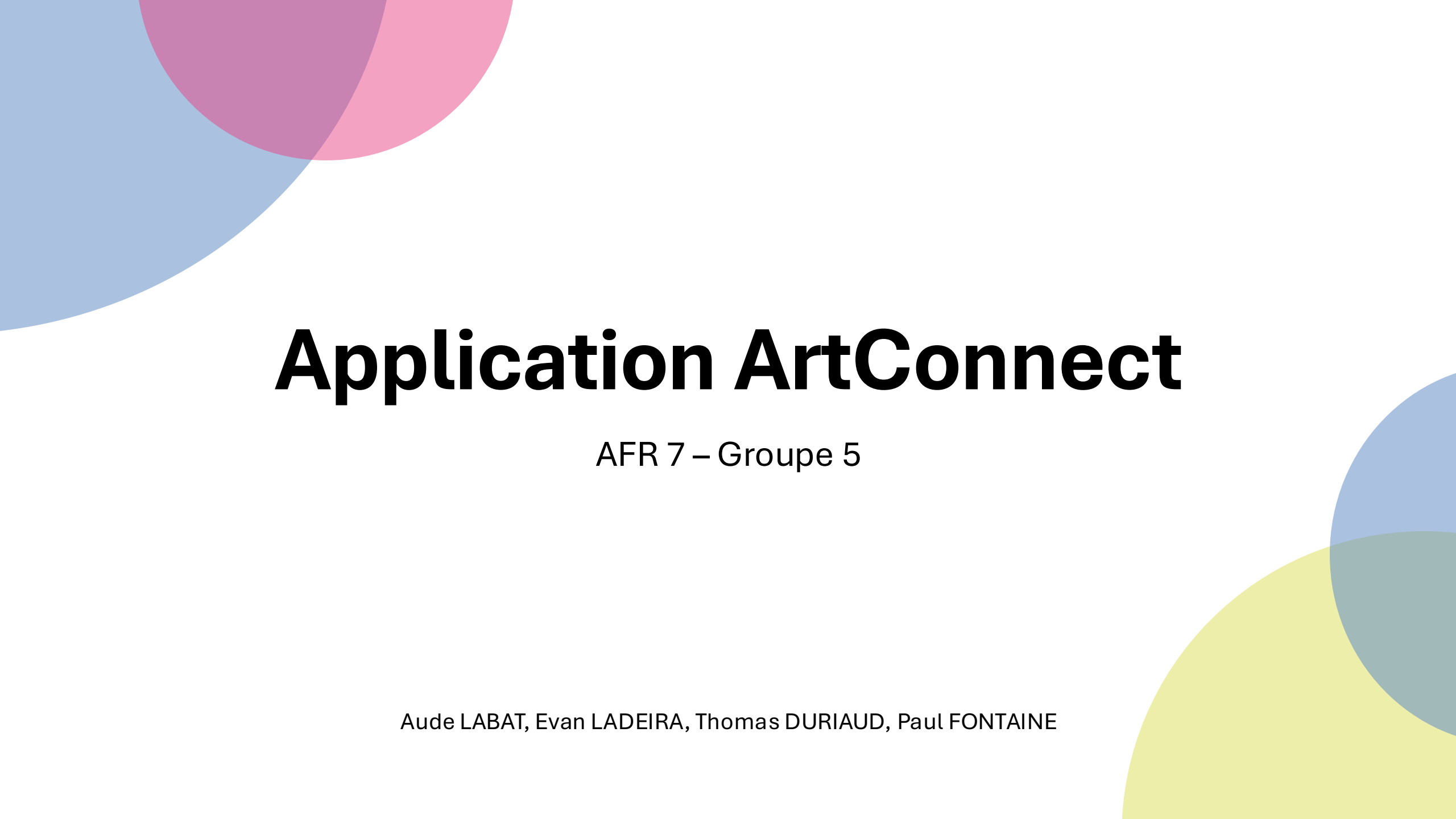




Application ArtConnect

AFR 7 – Groupe 5

Aude LABAT, Evan LADEIRA, Thomas DURIAUD, Paul FONTAINE

Introduction

→ Gérer le monde artistique :

Communauté

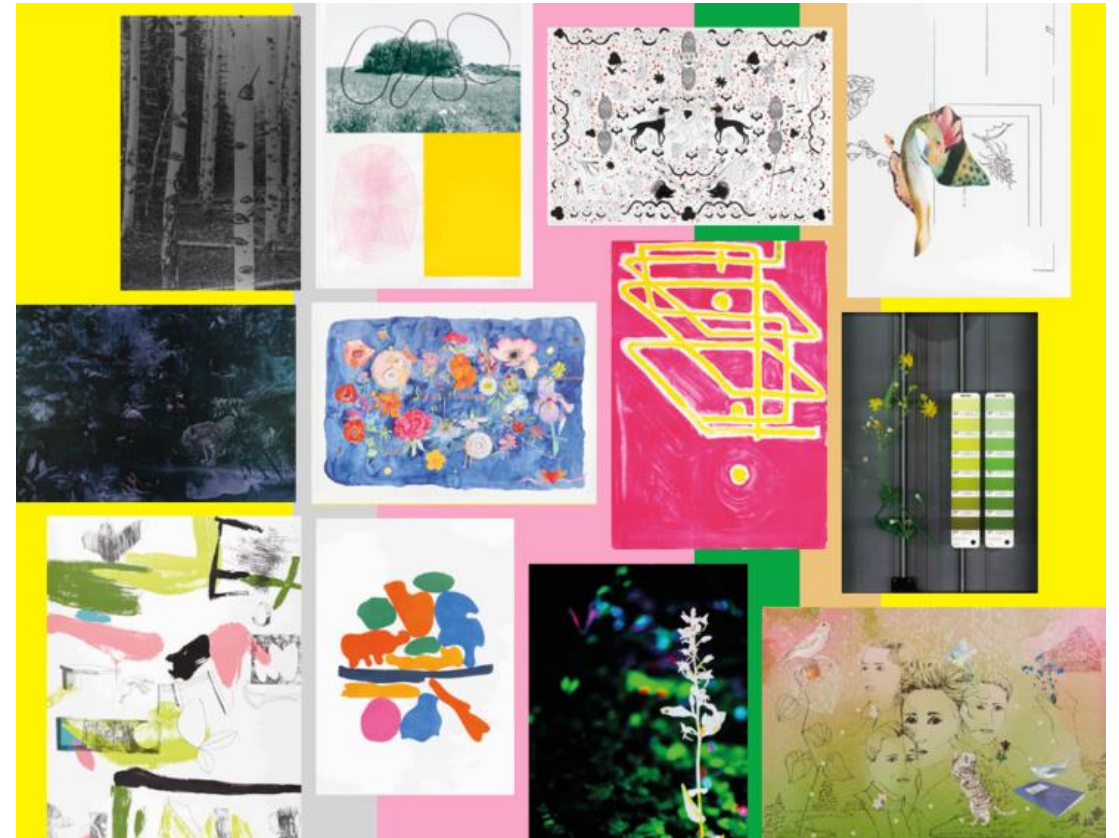
Ateliers

Artistes

Galleries

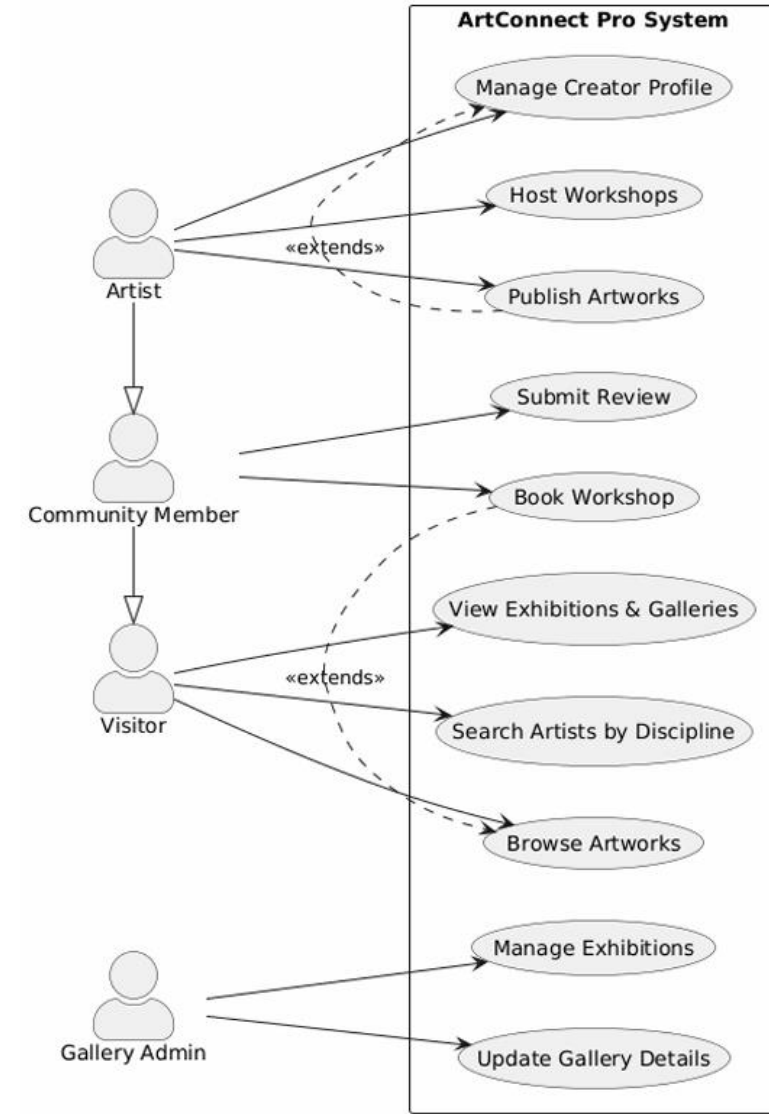
Expositions

Œuvres



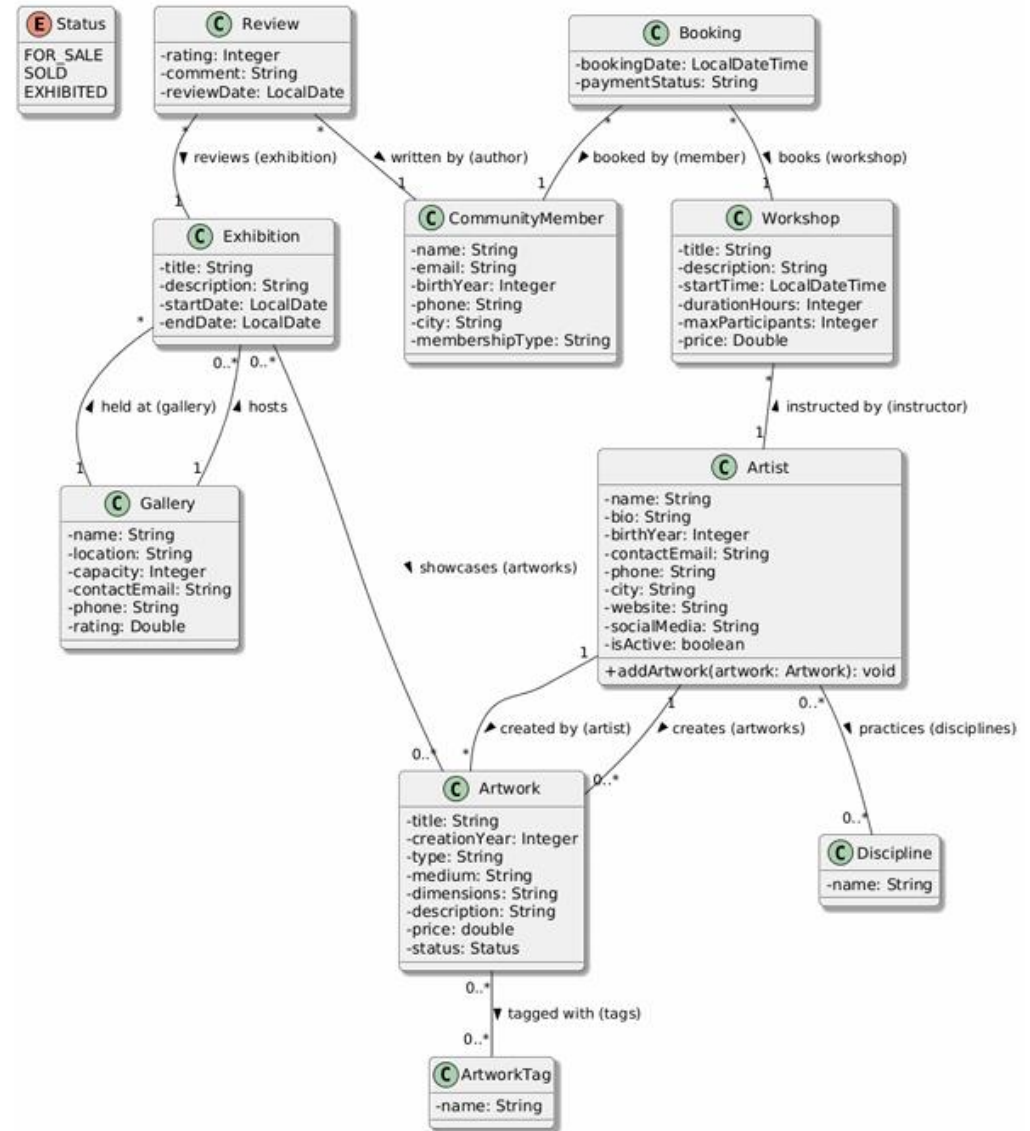
Analyse fonctionnelle

Diagramme des cas d'utilisation



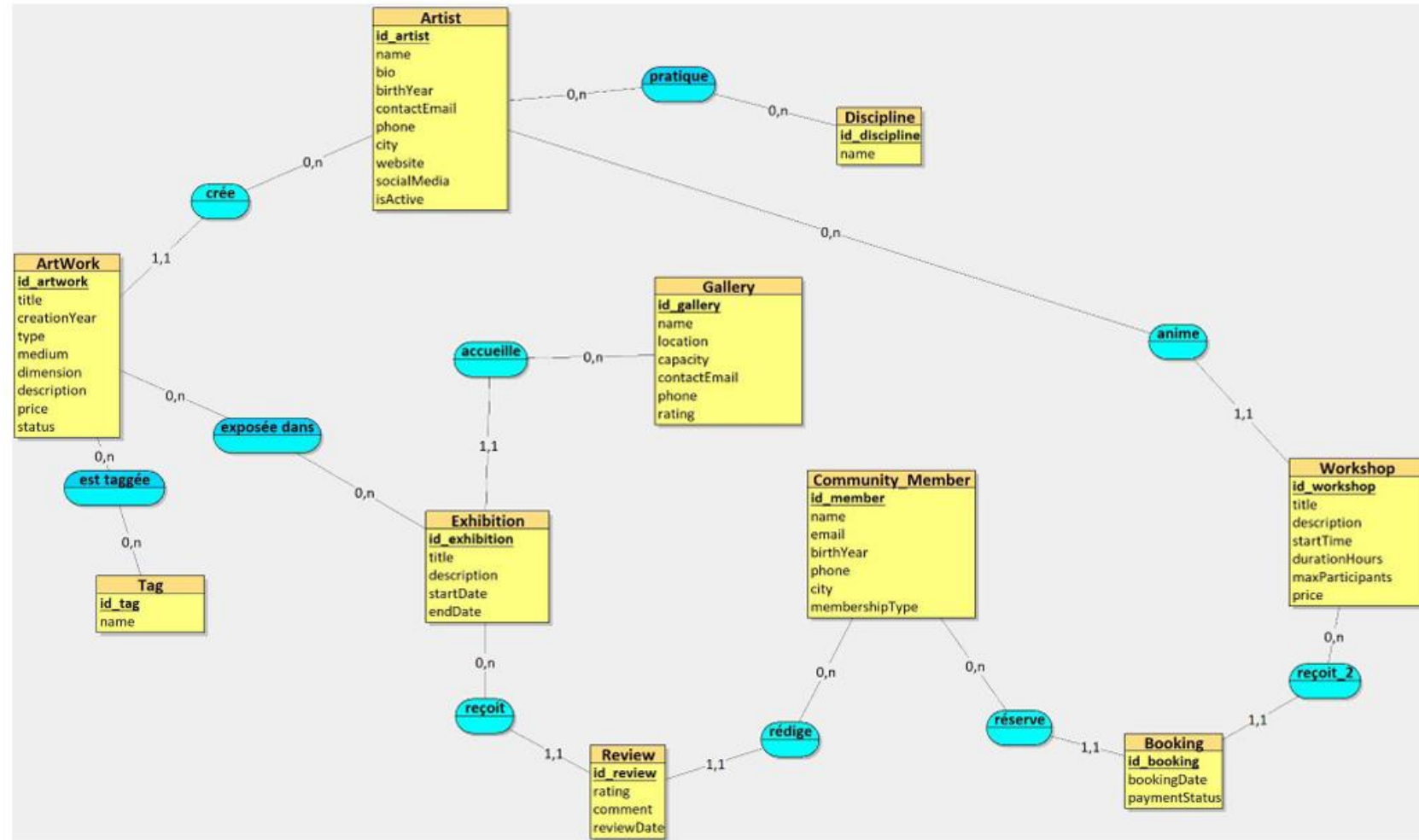
Analyse fonctionnelle

Diagramme de classe



Construction de la BDD

Modèle conceptuel de données (MCD)



Construction de la BDD

Modèle logique de données (MLD)

```
MLD
Artist = (id_artist INT, name VARCHAR(50), bio TEXT, birthYear INT, contactEmail VARCHAR(50), phone VARCHAR(50), city VARCHAR(50), website VARCHAR(50), socialMedia VARCHAR(50), isActive LOGICAL);
ArtWork = (id_artwork INT, title VARCHAR(50), creationYear INT, type VARCHAR(50), medium VARCHAR(50), dimension VARCHAR(50), description TEXT, price VARCHAR(50), status VARCHAR(50), #id_artist);
Tag = (id_tag INT, name VARCHAR(50));
Gallery = (id_gallery INT, name VARCHAR(50), location VARCHAR(50), capacity INT, contactEmail VARCHAR(50), phone VARCHAR(50), rating DECIMAL(15,2));
Workshop = (id_workshop INT, title VARCHAR(50), description VARCHAR(200), startTime DATE, durationHours INT, maxParticipants INT, price DECIMAL(15,2), #id_artist);
Community_Member = (id_member INT, name VARCHAR(50), email VARCHAR(50), birthYear INT, phone VARCHAR(50), city VARCHAR(50), membershipType VARCHAR(50));
Discipline = (id_discipline INT, name VARCHAR(50));
Exhibition = (id_exhibition INT, title VARCHAR(50), description VARCHAR(200), startDate DATE, endDate DATE, #id_gallery);
Booking = (id_booking INT, bookingDate DATE, paymentStatus VARCHAR(50), #id_workshop, #id_member);
Review = (id_review INT, rating INT, comment TEXT, reviewDate DATE, #id_member, #id_exhibition);
est_taggée = (#id_artwork, #id_tag);
exposée_dans = (#id_artwork, #id_exhibition);
pratique = (#id_artist, #id_discipline);
```


Construction de la BDD

Adaptation du MLD en SQL et création de la BDD

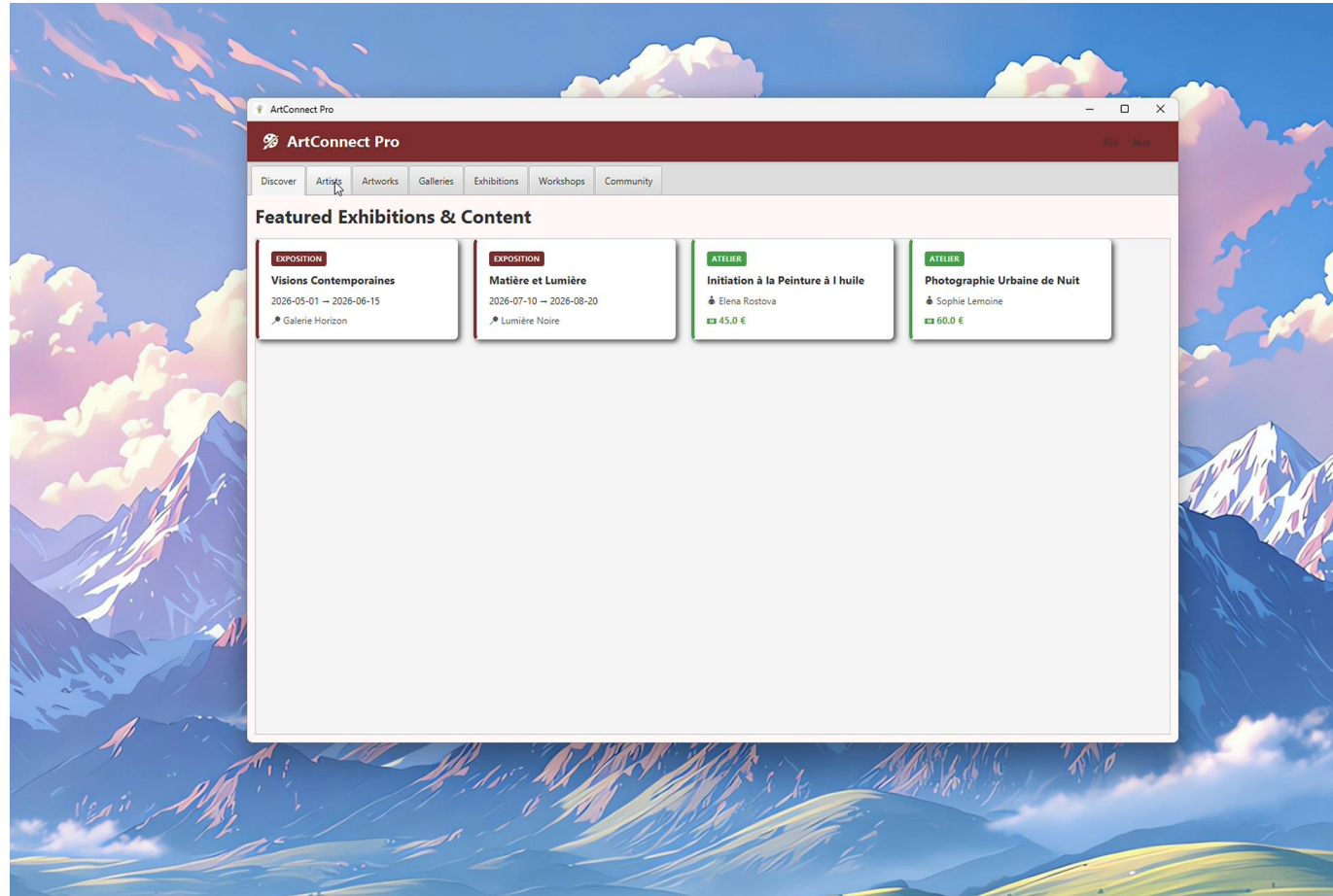
The screenshot displays the MySQL Workbench interface for a local instance. The left sidebar shows a list of schemas, including 'ecole', 'empreinte_carbone', 'smartbank', 'sys', 'tp2sql', 'tpsql', and 'voyages_extraordinaires'. The main editor window contains the following SQL queries:

```
6 DROP DATABASE IF EXISTS ArtConnect;
7 CREATE DATABASE ArtConnect;
8
9 USE ArtConnect;
10
11 CREATE TABLE Artist (
12     artist_id INT AUTO_INCREMENT PRIMARY KEY,
13     name VARCHAR(100) NOT NULL,
14     bio TEXT,
15     birthYear INT,
16     contactEmail VARCHAR(100),
17     phone VARCHAR(30),
18     city VARCHAR(100),
19     website VARCHAR(255),
20     socialMedia VARCHAR(255),
21     isActive BOOLEAN
22 );
23
24 CREATE TABLE Discipline (
25     discipline_id INT AUTO_INCREMENT PRIMARY KEY,
26     name VARCHAR(100) NOT NULL
27 );
28
29 CREATE TABLE Artist_Discipline (
30     artist id INT,
```

The bottom panel shows the 'Output' window with the 'Action Output' tab selected. It displays a log of executed queries and their results:

#	Time	Action	Message	Duration / Fetch
51	17:00:07	CREATE TABLE Artwork_Tag (artwork_id INT, tag_id INT, PRIMARY KEY (artwork_id, tag_id), FORE...	0 row(s) affected	0.062 sec
52	17:00:07	CREATE TABLE Gallery (gallery_id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(150), lo...	0 row(s) affected	0.047 sec
53	17:00:07	CREATE TABLE Exhibition (exhibition_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200)...	0 row(s) affected	0.063 sec
54	17:00:07	CREATE TABLE Artwork_Exhibition (artwork_id INT, exhibition_id INT, PRIMARY KEY (artwork_id, exhi...	0 row(s) affected	0.078 sec
55	17:00:07	CREATE TABLE CommunityMember (member_id INT AUTO_INCREMENT PRIMARY KEY, name VARC...	0 row(s) affected	0.047 sec
56	17:00:07	CREATE TABLE Review (review_id INT AUTO_INCREMENT PRIMARY KEY, rating INT CHECK (rating ...	0 row(s) affected	0.078 sec
57	17:00:07	CREATE TABLE Workshop (workshop_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200)...	0 row(s) affected	0.062 sec
58	17:00:07	CREATE TABLE Booking (booking_id INT AUTO_INCREMENT PRIMARY KEY, bookingDate DATETI...	0 row(s) affected	0.078 sec

Démonstration de l'application



Démonstration du SQL

```
SELECT ar.name, d.name AS discipline FROM Artist ar JOIN Artist_Discipline ad ON  
ar.artist_id = ad.artist_id JOIN Discipline d ON d.discipline_id = ad.discipline_id  
WHERE ar.isActive = TRUE;
```

name	discipline
Elena Rostova	Peinture
Marcus Dubois	Sculpture
Sophie Lemoine	Photographie

```
SELECT FN_Get_Remaining_Seats(1) AS places_restantes;
```

places_restantes
7

```
INSERT INTO Exhibition (title, startDate, endDate, gallery_id)  
VALUES ('Test', '2026-10-01', '2026-09-01', 1);
```

```
EXPLAIN SELECT * FROM Artist WHERE name = 'Elena Rostova';
```

MySQL a répondu : ?

#1644 - La date de fin ne peut pas être antérieure à la date de début.

id	select_type	table	type	possible_keys	key	key_len	ref	rows	Extra
1	SIMPLE	Artist	ref	idx_artist_name	idx_artist_name	402	const	1	Using index condition

Bilan

